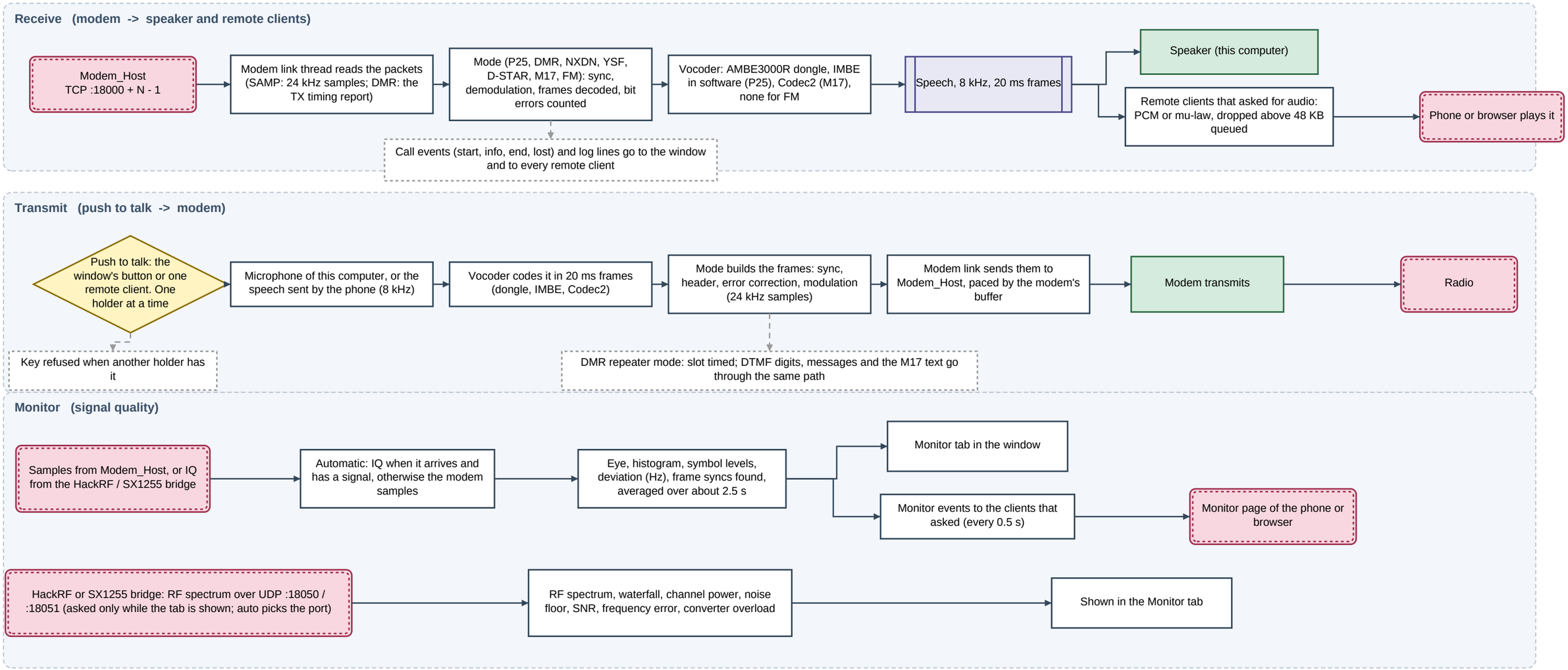
OpenMT Radio - the program

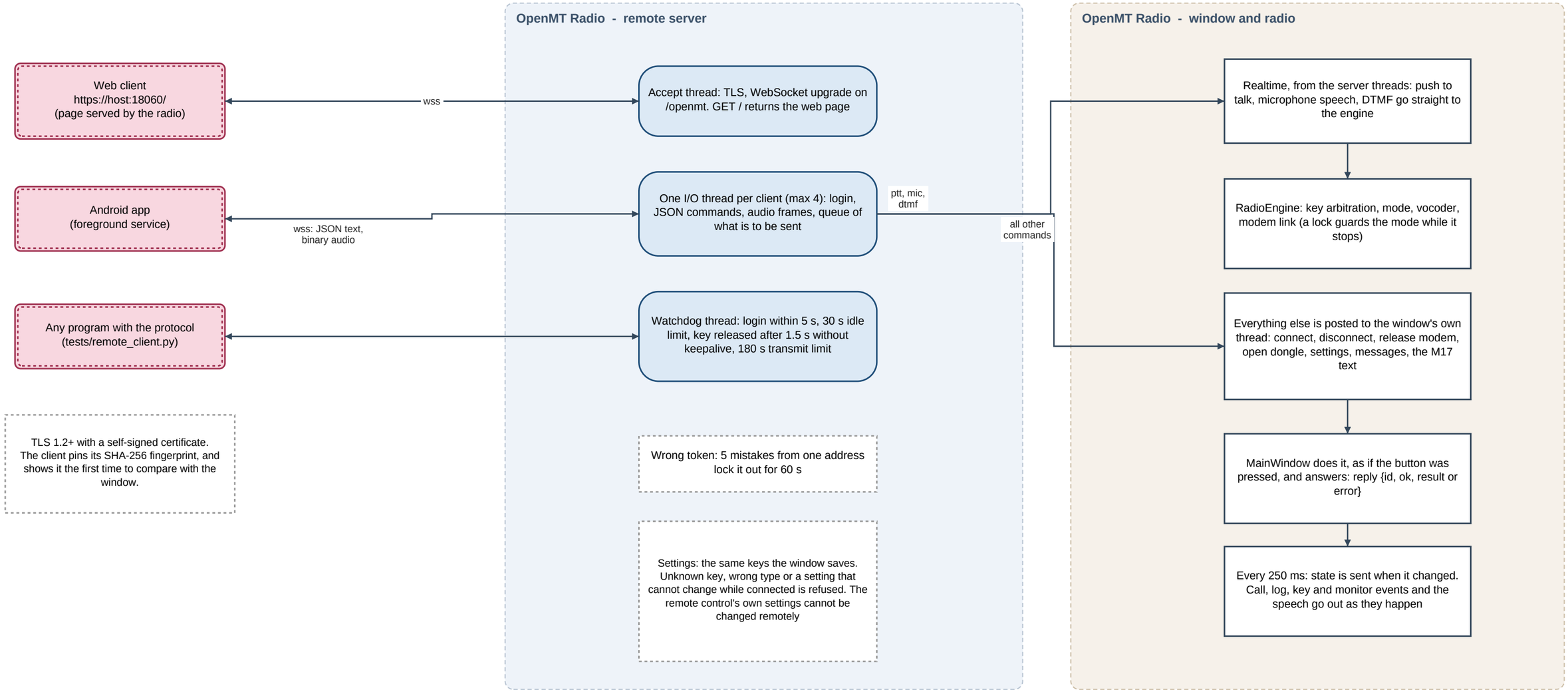
A radio on the modem: how speech and signals go in and out, and where the remote control, the web client and the Android app fit.



One OpenMT Radio per modem; it takes the modem with a mode of its own and gives it back when the radio is released (Release modem, in the window or from a remote client).
The window, the web client and the Android app all use the same commands and settings; the levels of the speaker and microphone on a phone are the phone's own.

OpenMT Radio - remote control

Who talks to whom when a phone or browser controls the radio.

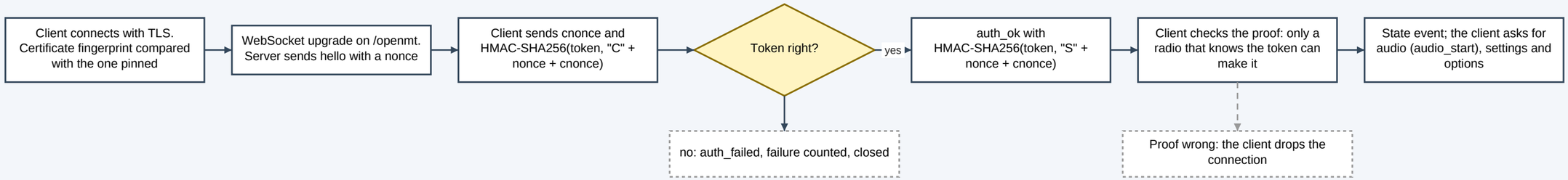


Everything the window does can be done remotely. The token (24 characters, shown in the window) is never sent: client and radio prove to each other that they know it.
The protocol is in clients/openmt_radio/REMOTE_PROTOCOL.md; the web client is clients/openmt_radio/web/index.html; the Android app is in clients/openmt_radio/android.

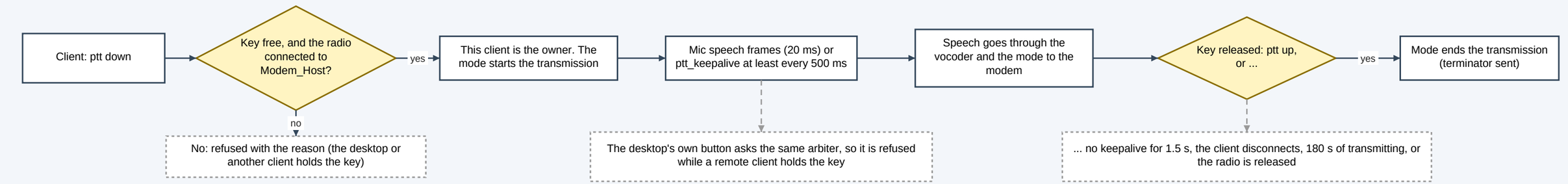
OpenMT Radio - login, push to talk and commands

What happens when a client connects, takes the key and sends commands.

Login



Push to talk (the transmitter never waits for the window)



A command

Audio frames: byte 0 type (1 = from the radio, 2 = from the client), byte 1 codec (0 PCM 16 bit, 1 mu-law), bytes 2-3 sequence, then 160 samples at 8 kHz.
Clients that stop reading are dropped (audio frames are discarded above 48 KB queued, the client above 2 MB).